

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 7 - Summative Assessment 2

Summative Assessment (SA-2) - Answer Key

Class: Class 7

Assessment Type: Summative Assessment (SA-2)

Total Marks: 40

Duration: 2 hours

Topics Covered: Ch 1-6: Complete Syllabus

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Part A: Multiple Choice Questions (10 marks)

Q.No	Answer	Explanation
1	c) Sound	The Sound category contains blocks for playing sounds and controlling volume.
2	b) Creative Commons	CC stands for Creative Commons, a licensing framework for sharing creative works.
3	b) POP3	POP3 (Post Office Protocol 3) is used to receive emails from a mail server.
4	b) Gaussian Blur	Gaussian Blur is a filter that creates a smooth blur effect on images.
5	a) Find the total of a range of cells	The SUM function adds all numeric values in a specified range.
6	b) The trail of data left by a user's online activities	A digital footprint is the record of all online activities.
7	a) create clone of myself	This block creates a duplicate copy of the current sprite.
8	d) Tree Topology	Star, Bus, and Ring are common topologies. Tree is a hierarchical structure, not a basic topology type.
9	b) Stops all scripts in the project	The stop all block halts every running script in the entire project.
10	c) Keeping software and antivirus updated	Updating software and antivirus is a safe online practice that protects against threats.

Part B: Short Answer Questions (16 marks)

1. What is the difference between a client and a server in a network?

- **Answer:** A client is a computer that requests services (e.g., browsing a website). A server is a computer that provides services to clients (e.g., hosting a website).
- **Marking:** 1 mark for client, 1 mark for server.

2. Explain any two Internet services with their uses.

- **Answer:** 1) Email — for sending and receiving messages electronically. 2) FTP — for transferring files between computers over the Internet.
- **Marking:** 1 mark for each service with use.

3. What are layers in GIMP? How do they help in editing images?

- **Answer:** Layers are transparent sheets for organizing image elements. They help by allowing non-destructive editing, editing individual parts without affecting others, and easy rearrangement.

- **Marking:** 1 mark for definition, 1 mark for explaining benefits.
4. **What is the difference between JPEG and PNG image formats?**
 - **Answer:** JPEG uses lossy compression, smaller file size, good for photographs. PNG uses lossless compression, supports transparency, good for graphics and logos.
 - **Marking:** 1 mark for JPEG, 1 mark for PNG.
 5. **Explain the concept of animation. What are the two main types of animation?**
 - **Answer:** Animation creates the illusion of movement using sequences of images. Two types: 1) Frame animation — individual images displayed in sequence. 2) Vector animation — mathematical paths define movement.
 - **Marking:** 1 mark for concept, 1 mark for types.
 6. **What is a formula in a spreadsheet? Write the syntax of the AVERAGE function.**
 - **Answer:** A formula is a calculation expression in a spreadsheet. Syntax: =AVERAGE (range).
Example: =AVERAGE (A1:A10).
 - **Marking:** 1 mark for definition, 1 mark for syntax.
 7. **What are “clones” in Scratch? How are they useful in game development?**
 - **Answer:** Clones are duplicate sprites created during runtime. They are useful for creating multiple similar objects (e.g., bullets, enemies, stars) without manually creating each one.
 - **Marking:** 1 mark for definition, 1 mark for usefulness.
 8. **What is plagiarism? Why should students avoid it?**
 - **Answer:** Plagiarism is using someone’s work without credit. Students should avoid it because it is dishonest, can result in academic penalties, and violates intellectual property rights.
 - **Marking:** 1 mark for definition, 1 mark for reasons to avoid.

Part C: Long Answer Questions (12 marks)

1. **Explain the complete anatomy of a computer network with a diagram. Include the role of routers, switches, and modems in a network.**
 - **Answer:**
 - **Routers:** Direct data packets between different networks.
 - **Switches:** Connect devices within a LAN and manage data flow.
 - **Modems:** Convert digital signals to analog for transmission over phone lines.
 - Diagram should show these components connected with proper labels.
 - **Marking:** 2 marks for component explanations, 1 mark for diagram.
2. **Describe the steps to create an animated story in Scratch. Include the use of at least four different block categories.**
 - **Answer:**
 - Step 1: Choose sprites and backgrounds.
 - Step 2: Use Events blocks to start the story.
 - Step 3: Use Looks blocks for speech bubbles and costumes.
 - Step 4: Use Motion blocks for sprite movement.
 - Step 5: Use Control blocks for timing and loops.
 - Step 6: Use Sound blocks for background music.
 - **Marking:** 2 marks for steps, 1 mark for block categories.
3. **What is data handling? Explain the steps to create a dashboard in LibreOffice Calc using charts and formulas. Give a practical example.**
 - **Answer:**
 - Data handling involves collecting, organizing, analyzing, and presenting data.

- Steps: 1) Enter data in cells. 2) Use formulas (SUM, AVERAGE) for calculations. 3) Select data range. 4) Insert chart (bar, pie, line). 5) Customize chart labels and colors.
 - Example: Student marks analysis with total, average, and grade charts.
 - **Marking:** 1 mark for data handling definition, 2 marks for steps and example.
4. **Explain the importance of digital citizenship. Describe any five rules of responsible technology use with examples.**
- **Answer:**
 - Digital citizenship means using technology responsibly and ethically.
 - Rules: 1) Protect personal information online. 2) Think before you post. 3) Respect others online. 4) Do not plagiarize. 5) Report cyberbullying.
 - **Marking:** 1 mark for importance, 0.4 marks for each rule (2 marks for five rules).

Part D: Application Based Questions (2 marks)

1. **Meera wants to create a presentation with background music and animated images. Which multimedia tools would you recommend? Explain how she can use them.**
- **Answer:** Recommend: 1) LibreOffice Impress for presentation creation. 2) Audacity for editing background music. 3) GIMP for editing animated images. She can add audio files to slides and create animated GIFs.
 - **Marking:** 1 mark for tool recommendations, 1 mark for usage explanation.
2. **A student receives an email asking for their bank details with a link to “update account information”. What should the student do and why?**
- **Answer:** The student should NOT click the link or provide any information. This is a phishing attempt. The student should report the email to a teacher or parent and delete it immediately.
 - **Marking:** 1 mark for correct action, 1 mark for explaining why.